

ELEC362 - Week 9

Practical Exercise

1. Re-implement the GUI application made in Lecture 14. As a reminder, the application allows the user to type a paragraph and to type a keyword, the application counts how any times the word is repeated in the paragraph and displays to the user the number of counts. Ask for help if you are unsure what to do.
2. Verify the application works correctly by typing the sentence “Hello, how are you?” then count the occurrences of “Hello”, then “how”, what does the counter show? Why? Fix that using two different ways (Hint: one of them is a more suitable function for the QLineEdit class).
3. Add a “clear” button to the app that clears everything when it is pressed.
4. Type the sentence “he said hello” then count the occurrences of “he”. The app will return 2 as it counts every occurrence whether it is part of a word or not. Write your own function that only counts the number of matching words (i.e. for input “he said hello”, the word “he” will have 1 occurrence only).

Hint: Write the function as you would do in standard C++, QString is similar to standard strings in C++

5. Implement a counter GUI application that enables the user to set the time they want to wait for. The app should have “start”, “stop” and “reset” buttons. The countdown starts from the number the user sets to 0, it starts when the user pushes the “start” button. You get to decide how the user is going to input the countdown time.
6. When the user pushes “stop” the countdown should stop, when the user pushes “reset” the counter should read whatever, the user has input as the new value of the counter.
7. Make sure the app’s user interface is well formatted.

Hint: To control time, you need an object of the class “QTimer”. <https://doc.qt.io/qt-6/qtimer.html>